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Rootkit Analysis

Rootkits are a relevant factor in the world of cybersecurity. They are software packages that can be difficult to detect and can be used to compromise certain operating Systems. The term “root” refers back to administrative permissions, while ‘kit’ can stand for the tools that are brought along. Rootkits can be considered a type of malware while being used by attackers, although this is not always the case. Because of this, a rootkit attack may also be a usable term, as it is simply the attacker using a rootkit as the means of compromise. The main goal of utilizing a rootkit is to reach the ‘root’ or ‘kernel’ of the program, as this also provides an anonymous presence to the owner (Beegle, 2007). Tools like stealth are a staple for rootkits when used by attackers. This means some rootkit attacks can be invisible to antivirus programs. Since invisibility is nothing new to the security industry, you could consider these situations a malware war, with ever evolving technologies on both sides (Eset, 2025). Tools like this have kept rootkit a relevant threat against Operating systems and system security for years and it is critical to remain knowledgeable on how they work in the security world. This paper aims to show how rootkits function, methods used for attacks, and the prevention being developed to fight against rootkits .

It is common for rootkits to be introduced through malicious software such as viruses, ransomware, malware, and other keylogger programs. Rootkits can open a backdoor on a victim's system to execute an attack (Robinson, 2023). But what exactly can a rootkit access once it has attempted its goal? Rootkits can modify components and access unauthorized files. The rootkit would also be able to access things like hidden threads, folders, directories,

credentials, subdirectories, and open ports (Eset, 2025). The Rootkit would also aim to hide any presence of itself and activities done to the host system.

The history of rootkits is considered to have formed sometime in the 90's, with some tools being established prior, such as file cleaners. (Chuvakin 2003). Overtime rootkits have evolved. A great summary of rootkit functionality can be summarized as maintaining access, attacking systems, destroying evidence (Chuvakin 2003). Rootkits have evolved heavily throughout the years, and it is essential to look at its full history, along with the different types of kits. One famous example of a rootkit incident occurred in 2005 when Sony had installed rootkits on over 25 million CD's. This ended up installing digital rights management software on the consumers device which resulted in modifying their OS and leaving them vulnerable to other forms of malware, leading to class action lawsuits (Robinson, 2023). The rootkit in this situation modified the OS where it prevented all files beginning with a certain prefix from being revealed, and gave its own files those prefixes (Gross, 2016).

Common types of rootkits would be user-mode and kernel-mode. User-mode can run within a user application. Because of this, user-mode can be easier to detect, while kernel-mode rootkits can run deeper within the OS with more potential for damage. (Eset, 2025). Since kernel-mode runs within a systems kernel, things like data structures can also be tampered with and modified, while also bypassing security. This means kernel-mode can gain much more control and manipulation of a system, all while being harder to detect.

It can be difficult to prevent a rootkit attack depending on the situation, but many protective measures do exist to keep you safe and aware of any suspicious activity. The most first and most obvious measure would be antivirus/antimalware software. This still may require the

host to know what they're doing, as add-ons may be required as well. Specific add-ons such as special purpose anti-rootkit scanner software would be an example (Robinson, 2023). For a modern Linux OS, Kernel lockdown mode can be implemented, which helps limit access to things like kernel memories, and makes things harder for rootkits to take control. Another protective term is Secure boot, which plays a security role for Windows OS. Security against rootkits also leads to more secure detections. Some of the methods are memory dumps, which allow analysis of a ram when needed, Integrity checks are also another practice to help detect the presence of a rootkit. “It’s apparent that rootkits pose a formidable threat, and not just because they’re sometimes installed by vendors (like Sony) or created by bad actors (like hackers). They are swiftly evolving in incredible ways as researchers break new ground” (Gross, 2016).

Since rootkits are hard to detect, several more tools have been developed over the years to recognize subtle symptoms a rootkit may leave behind, such as changes to systems memory or behavior. Chkrootkit is a tool used to detect Linux systems. More specifically, it is a command-line tool that checks for signs of a rootkit on your server, scans binaries, and checks for hidden connections (wafatech, 2025). Another tool goes by the name of Lynis. Lynis runs on Linux, MacOS, and Unix systems. Lynis goes more in depth in terms of checking for systems securities. Having given these protective tools, it is important to keep things simple and use a responsible updated OS, as stated by Malwarebytes. “The first line of defense is reducing the surface of attack by using a modern operating system that implements countermeasures against rootkits. For example, 64-bit editions of MS Windows require all kernel-level drivers to be signed in order to prevent unauthorized code from sneaking in” (Malwarebytes, 2025).

It is worth mentioning how each defense and technique differ depending on the OS they are running on. As mentioned previously, Linux OS may use security tools such as chkrootkit and lynus. A third tool is 'rkhunter'. "Rootkit Hunter (rkhunter) is a standard tool for the detection of rootkits on Linux. The Ubuntu Linux distribution maintains a package for rkhunter. The complete set of checks that rkhunter performs provides a good security baseline for finding some of the most malicious elements of the offensive security landscape. It is recommended you regularly run the rkhunter" (Extreme, 2022). As for Windows OS, tools like Microsoft defender may be able to detect any potential signs of rootkits, while offering protection in real time. Other third party tools like Malwarebytes is another security capable of running on Windows. So how effective are tools like these against modern and evolving rootkits? The answer can be tricky because even with these tools, it still may be difficult to find the rootkit because of how elusive they are within the systems.

For more hands-on experience and deeper understanding, one can have virtual labs for testing and experimentation, without harming themselves. Labs such as VMware and VirtualBox can allow users to run and experiment with their own rootkit. Simulating this in Virtual labs can create real world situations and provide more insight on how rootkits function, and is helpful in research.

In terms of virtual environments, there are real world issues that present themselves as well. "A suspected Chinese threat actor tracked as UNC3886 uses publicly available open-source rootkits named 'Reptile' and 'Medusa' to remain hidden on VMware ESXi virtual machines, allowing them to conduct credential theft, command execution, and lateral movement" (Toulas, 2024). This situation shows how the bad actor utilized vulnerabilities to compromise the

VMware EXSi, or virtual machines, and used rootkits named ‘Medusa’ and ‘Reptile’ to gain access and remain anonymous , ultimately achieving things like executing theft and compromising networks. ‘Reptile’ was first deployed, which contained user mode components which connected with kernel mode to things like hide files and connections (Toulas, 2024).

In conclusion, rootkits remain an expanding and progressing threat to systems and cybersecurity. From developing tools in the 1990’s, to more exposedured rootkit incidents in the 2000’s, it remains an evolving staple that is adapting with current operating systems, technologies, and anti-rootkit software. Understanding how rootkits function along with how and why they are used in attacks is ideal in today’s world.

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